

Recovery status memo

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Problem

- What is the current state of recovery of human body?
- Recovery is a combination of subjective and objective factors e.g.
 - heart rate data (HR + HRV)
 - subjective feeling
 - other indicators (lactic acid, hormone levels, ...)
- → Recovery can be thought as a latent state
- When recovery is monitored, a variety of different factors are affecting on the results
 - physical and mental stress, day of week, month, etc.
- Need to separate the sources of test condition related effects
 - No matter how good or bad your recovery status is, conditions are affecting on the measurements e.g.
 - * testing at warm conditions vs. cold conditions
 - * testing in the morning vs. evening
 - * testing at sea level vs. high altitude

Aims

- 1) Could we get comparable information about the recovery of an individual over time by estimating the counterfactuals of test results?
- 2) Could we use these estimates as observations and subjective feeling as a prior information for detecting the latent state of recovery?
- 3) Could we use these latent states (estimated over time) to define a homeostasis (range) of recovery?
- 4) Could we determine the recovery status by comparing the current state of recovery with the homeostasis?

Variables

- Y = response
 - testing heart rate levels at different loads on a treadmill
 - * response following Multinormal distribution
 - (nocturnal HRV as possible additional data)***
- X = test protocol
 - speed and angle of the treadmill
 - duration spent at each load
- C = test conditions
 - time of day, heat, humidity, altitude ...
- Z = other factors affecting
 - physical and mental stress, month, ...

Test protocol

- ‘Little treadmill test’
 - test developed already at the early 1990s by the experts of cross-country ski
- Example of the test in practice

- 4x4min at xx km/h and yy angle
- with each phase the speed is kept constant but the angle is increased
- HR monitored during the test
 - * avg HR of the last 30 sec of each load and the recovery HR measured 1min after the last load
- (with professional athletes the test may also include lactic acid collection during the test)

Example data snapshot

##	id	time	HR	speed	angle	temperature	time_of_day	age	month	training_load
## 1	1	1	136	12	0	16	10	29	January	hard
## 2	1	1	148	12	2	16	10	29	January	hard
## 3	1	1	162	12	4	16	10	29	January	hard
## 4	1	1	171	12	6	16	10	29	January	hard
## 5	1	1	144	0	0	16	10	29	January	hard
## 6	1	8	133	12	0	15	19	29	January	light
## 7	1	8	143	12	2	15	19	29	January	light
## 8	1	8	160	12	4	15	19	29	January	light
## 9	1	8	170	12	6	15	19	29	January	light
## 10	1	8	145	0	0	15	19	29	January	light
## 11	...	...	...	...	...	...	...	...	...	...
## 12	2	1	130	12	0	16.5	11	32	January	light
## 13	2	1	144	12	2	16.5	11	32	January	light
## 14	2	1	158	12	4	16.5	11	32	January	light
## 15	2	1	165	12	6	16.5	11	32	January	light
## 16	2	1	130	0	0	16.5	11	32	January	light
## 17	2	10	133	12.5	0	15.5	16	32	January	hard
## 18	2	10	140	12.5	2	15.5	16	32	January	hard
## 19	2	10	160	12.5	4	15.5	16	32	January	hard
## 20	2	10	166	12.5	6	15.5	16	32	January	hard
## 21	2	10	137	0	0	15.5	16	32	January	hard

Estimation strategy

- 1) Estimate expected ‘condition standardized’ responses ($\hat{y}$)
 - $\hat{y} = E(y|do(x), do(c))$
- 2) Use $\hat{y}$ as observations to estimate the latent state of recovery θ
 - linear Gaussian State space model
 - $\hat{y}_t = Z_t \theta_t + \epsilon_t$
 - $\theta_{t+1} = T_t \theta_t + R_t \eta_t$
 - $t = 1, \dots, n$
 - $\epsilon_t \sim N(0, H_t)$
 - $\eta_t \sim N(0, Q_t)$
- 3) Define the homeostasis by setting the ranges of normal variation (how?)
 - $\Theta_t = \text{mean}(\theta_{t-k}, \theta_{t-k+1}, \dots, \theta_t) \pm \text{sd}(\theta_{t-k}, \theta_{t-k+1}, \dots, \theta_t)$
 - where $k \geq 0$
- 4) Derive the state of recovery
 - depending how ‘far’ the current state is from the mean of homeostasis
 - $|\Theta_t - \theta_t|$
 - within normal range: “recovered”
 - out of the normal range: “fatigued”
 - and possible classifications between

Questions and observations

- Can we take the subjective feeling of the recovery into account as a prior distribution of θ ?
 - uniform distr. as uninformative prior
 - normal distr. as informative prior
 - * strong prior would emphasize the role of subjective feeling about the state, but that would be totally fine $\rightarrow$ modifying the prior according how strong the subjective feeling is
- Can we combine multiple objective measures*** for defining the latent state?
 - HRV (1) can be monitored daily and Treadmill test (2) is run once a week (maximum)
 - * These produce different types of data $\rightarrow$ (1) can be thought as dependent observations but (2) not necessarily?
 - * $\rightarrow$ should we then somehow transform the data sources (1) and (2) to same scale by e.g. calculating averages of (1) to get weekly data as it is with (2)?